Spotify song popularity analysis

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```r
library(tidyverse)
library(rio)
library(olsrr)
library(broom)
library(blorr)
library(forecast)
```

---

1 Spotify

Data source:

https://www.kaggle.com/datasets/melissamonfared/spotify-tracks-attributes-and-popularity

1.1 Executive summary

This analysis examines how audio features influence song popularity on Spotify. Results show that more danceable and louder tracks tend to be more popular, whereas features such as instrumentalness and energy are associated with lower popularity.

Genre plays an important role. Some relationships, such as the positive effect of danceability and loudness, were consistent across genres, while others, including energy and duration, vary by music type. This suggests that popularity depends not only on audio characteristics but also on genre-specific dynamics.

Overall, audio features provide useful insight but explain only part of popularity, indicating that external factors such as artist recognition or marketing likely play a significant role.

1.2 Introduction

The Spotify Tracks Dataset [https://www.kaggle.com/datasets/melissamonfared/spotify-tracks-attributes-and-popularity] contains 114,000 observations across 21 variables, including track, artist, genre, popularity, and audio features derived from the Spotify API. The dataset appears artificially balanced, with roughly 1,000 observations per genre, although the sampling process is not documented.

The primary objective is to identify which audio variables best explain variation in popularity. A secondary objective is to assess whether these relationships vary by genre.

1.3 Data preparation

Data were inspected and variables grouped into:

- information variables (excluded),
- categorical variables (inspected but not used),
- audio variables (numeric predictors),
- target variable (popularity)

Observations with zero popularity were removed. Genres were grouped into broader categories (Pop, HipHop, Rock, Metal, Electronic, Classical) based on key terms, excluding unmatched tracks. Tracks may appear multiple times with different genre labels; therefore, observations were collapsed to one row per track to ensure independence and avoid duplication. Where multiple genre assignments existed, a single genre was retained, introducing a simplifying assumption that may not fully capture the multi-genre nature of some tracks. Only the target, genre, and numeric audio variables were retained, with numeric variables standardised.

The final sample size was n = 28,970 observations.

```r
spotify <- import("data/dataset.csv")
```

```r
# Create factors where appropriate
spotify$track_genre <- as.factor(spotify$track_genre)
spotify$key <- as.factor(spotify$key)
spotify$mode <- as.factor(spotify$mode)
spotify$time_signature <- as.factor(spotify$time_signature)

# Datatypes
str(spotify)
```

```
'data.frame':	114000 obs. of  21 variables:
 $ index           : int  0 1 2 3 4 5 6 7 8 9 ...
 $ track_id        : chr  "5SuOikwiRyPMVoIQDJUgSV" "4qPNDBW1i3p13qLCt0Ki3A" "1iJBSr7s
 $ artists         : chr  "Gen Hoshino" "Ben Woodward" "Ingrid Michaelson;ZAYN" "Kina
 $ album_name      : chr  "Comedy" "Ghost (Acoustic)" "To Begin Again" "Crazy Rich As
 $ track_name      : chr  "Comedy" "Ghost - Acoustic" "To Begin Again" "Can't Help Fa
 $ popularity      : int  73 55 57 71 82 58 74 80 74 56 ...
 $ duration_ms     : int  230666 149610 210826 201933 198853 214240 229400 242946 189
 $ explicit        : logi  FALSE FALSE FALSE FALSE FALSE FALSE ...
 $ danceability    : num  0.676 0.42 0.438 0.266 0.618 0.688 0.407 0.703 0.625 0.442
 $ energy          : num  0.461 0.166 0.359 0.0596 0.443 0.481 0.147 0.444 0.414 0.63
 $ key             : Factor w/ 12 levels "0","1","2","3",..: 2 2 1 1 3 7 3 12 1 2 ...
 $ loudness        : num  -6.75 -17.23 -9.73 -18.52 -9.68 ...
 $ mode            : Factor w/ 2 levels "0","1": 1 2 2 2 2 2 2 2 2 2 ...
 $ speechiness     : num  0.143 0.0763 0.0557 0.0363 0.0526 0.105 0.0355 0.0417 0.036
 $ acousticness    : num  0.0322 0.924 0.21 0.905 0.469 0.289 0.857 0.559 0.294 0.426
 $ instrumentalness: num  1.01e-06 5.56e-06 0.00 7.07e-05 0.00 0.00 2.89e-
06 0.00 0.00 4.19e-03 ...
 $ liveness        : num  0.358 0.101 0.117 0.132 0.0829 0.189 0.0913 0.0973 0.151 0.
 $ valence         : num  0.715 0.267 0.12 0.143 0.167 0.666 0.0765 0.712 0.669 0.196
 $ tempo           : num  87.9 77.5 76.3 181.7 119.9 ...
 $ time_signature  : Factor w/ 5 levels "0","1","3","4",..: 4 4 4 3 4 4 3 4 4 4 ...
 $ track_genre     : Factor w/ 114 levels "acoustic","afrobeat",..: 1 1 1 1 1 1 1 1 1
```

```r
# create variable groups for easier manipulation
info_vars <- c(
  "index", "track_id", "artists",
  "album_name", "track_name"
)

other_vars <- c(
  "explicit", "key", "mode",
  "time_signature"
)

audio_vars <- c(
  "duration_ms", "danceability", "energy",
  "loudness", "speechiness", "acousticness",
  "instrumentalness", "liveness",
  "valence", "tempo"
)

target <- "popularity"
```

```r
# Distribution
summary(spotify[, other_vars])
```

```
 explicit              key          mode        time_signature
Mode :logical    7       :13245   0:41319    0:    163
FALSE:104253     0       :13061   1:72681    1:    973
TRUE :9747       2       :11644              3:   9195
                 9       :11313              4:101843
                 1       :10772              5:   1826
                 5       : 9368
                 (Other):44597
```

```r
summary(spotify[, audio_vars])
```

```
 duration_ms        danceability        energy            loudness
Min.   :      0   Min.   :0.0000   Min.   :0.0000   Min.   :-49.531
1st Qu.: 174066   1st Qu.:0.4560   1st Qu.:0.4720   1st Qu.:-10.013
Median : 212906   Median :0.5800   Median :0.6850   Median : -7.004
Mean   : 228029   Mean   :0.5668   Mean   :0.6414   Mean   : -8.259
3rd Qu.: 261506   3rd Qu.:0.6950   3rd Qu.:0.8540   3rd Qu.: -5.003
Max.   :5237295   Max.   :0.9850   Max.   :1.0000   Max.   :  4.532
 speechiness        acousticness     instrumentalness       liveness
Min.   :0.00000   Min.   :0.0000   Min.   :0.00e+00   Min.   :0.0000
1st Qu.:0.03590   1st Qu.:0.0169   1st Qu.:0.00e+00   1st Qu.:0.0980
Median :0.04890   Median :0.1690   Median :4.16e-05   Median :0.1320
Mean   :0.08465   Mean   :0.3149   Mean   :1.56e-01   Mean   :0.2136
3rd Qu.:0.08450   3rd Qu.:0.5980   3rd Qu.:4.90e-02   3rd Qu.:0.2730
Max.   :0.96500   Max.   :0.9960   Max.   :1.00e+00   Max.   :1.0000
```

```
   valence            tempo
 Min.   :0.0000   Min.   :  0.00
 1st Qu.:0.2600   1st Qu.: 99.22
 Median :0.4640   Median :122.02
 Mean   :0.4741   Mean   :122.15
 3rd Qu.:0.6830   3rd Qu.:140.07
 Max.   :0.9950   Max.   :243.37
```

```r
summary(spotify[, target])
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   17.00   35.00   33.24   50.00  100.00
```

```r
spotify <- spotify %>%

  # 1. Map genres
  mutate(
    genre = case_when(
      track_genre %in% c("pop","k-pop","j-pop","mandopop","cantopop",
                         "indie-pop","power-pop","synth-pop","pop-film") ~ "Pop",

      track_genre %in% c("hip-hop","r-n-b") ~ "HipHop",

      track_genre %in% c("rock","alt-rock","rock-n-roll","rockabilly",
                         "hard-rock","psych-rock","alternative","indie",
                         "grunge","emo","j-rock","punk-rock") ~ "Rock",

      track_genre %in% c("metal","heavy-metal","death-metal",
                         "black-metal","metalcore") ~ "Metal",

      track_genre %in% c("electronic","electro","edm",
                         "house","techno","trance",
                         "dubstep","drum-and-bass","breakbeat","trip-hop") ~ "Elect

      track_genre %in% c("classical","opera") ~ "Classical",

      TRUE ~ NA_character_
    )
  ) %>%

  # 2. Keep rows with genre identified
  filter(!is.na(genre), popularity != 0) %>%

  # 3. Collapse to one row per track (to fix duplication)
  group_by(track_id) %>%
  slice_head(n = 1) %>%
  ungroup() %>%
```

```r
  # Further processing
  dplyr::select(-all_of(c(info_vars, other_vars))) %>%
  mutate(across(all_of(audio_vars), scale)) %>%
  dplyr::select(-track_genre)
```

```r
spotify %>%
  count(track_id) %>%
  filter(n > 1)
```

```r
rio::export(spotify, "outputs/spotify_clean.xlsx")
```

```r
# genre - factor
spotify$genre <- as.factor(spotify$genre)

# Datatypes - check again
str(spotify)
```

```
tibble [28,970 x 12] (S3: tbl_df/tbl/data.frame)
 $ popularity      : int [1:28970] 62 38 20 86 36 58 36 16 44 33 ...
 $ duration_ms     : num [1:28970, 1] -0.462 -0.582 0.11 -0.16 -0.55 ...
  ..- attr(*, "scaled:center")= num 240386
  ..- attr(*, "scaled:scale")= num 108513
 $ danceability    : num [1:28970, 1] 0.848 0.0808 -0.7724 -1.159 -
0.9074 ...
  ..- attr(*, "scaled:center")= num 0.541
  ..- attr(*, "scaled:scale")= num 0.163
 $ energy          : num [1:28970, 1] 0.31 0.965 0.822 0.922 0.349 ...
  ..- attr(*, "scaled:center")= num 0.698
  ..- attr(*, "scaled:scale")= num 0.231
 $ loudness        : num [1:28970, 1] 0.941 0.679 1.189 0.519 -0.197 ...
  ..- attr(*, "scaled:center")= num -7.31
  ..- attr(*, "scaled:scale")= num 4.01
 $ speechiness     : num [1:28970, 1] 1.7406 0.0512 -0.0568 0.0349 -
0.5361 ...
  ..- attr(*, "scaled:center")= num 0.0723
  ..- attr(*, "scaled:scale")= num 0.0676
 $ acousticness    : num [1:28970, 1] -0.567 -0.697 -0.745 -0.759 -
0.753 ...
  ..- attr(*, "scaled:center")= num 0.227
  ..- attr(*, "scaled:scale")= num 0.298
 $ instrumentalness: num [1:28970, 1] -0.496 -0.1871 -0.489 -0.496 0.0685 ...
  ..- attr(*, "scaled:center")= num 0.141
  ..- attr(*, "scaled:scale")= num 0.285
 $ liveness        : num [1:28970, 1] -0.726 0.688 -0.132 -0.628 0.424 ...
  ..- attr(*, "scaled:center")= num 0.209
  ..- attr(*, "scaled:scale")= num 0.174
 $ valence         : num [1:28970, 1] 1.59 1.029 -1.207 -0.844 0.322 ...
```

```
  ..- attr(*, "scaled:center")= num 0.445
  ..- attr(*, "scaled:scale")= num 0.248
 $ tempo           : num [1:28970, 1] 1.22 1.955 1.666 0.759 -1.248 ...
  ..- attr(*, "scaled:center")= num 125
  ..- attr(*, "scaled:scale")= num 29.7
 $ genre           : Factor w/ 6 levels "Classical","Electronic",..: 3 6 2 6 6 6 6 2 5
```

```r
# final sample size
nrow(spotify)
```

```
[1] 28970
```

1.4 Part 1 - Global models

1.4.1 Multiple linear regression models

Three linear models were fitted:

- lm_no_genre: baseline without genre, only numeric audio variables
- lm_main: includes genre
- lm_interaction: includes genre interactions

Model performance improves with complexity, with adjusted R^2 increasing from approximately 0.09 (no genre) to 0.14 (with genre) and 0.18 (interaction model), (Table 1). This indicates that genre contributes meaningfully and that relationships vary across genres, even as overall explanatory power was limited.

```r
data = spotify

# fit linear models
lm_no_genre <- lm(popularity ~ . - genre, data = data)
lm_main <- lm(popularity ~ ., data = data)
lm_interaction <- lm(popularity ~ . * genre, data = data)
```

```r
no_genre    <- t(glance(lm_no_genre))
main        <- t(glance(lm_main))
interaction <- t(glance(lm_interaction))

model_table <- cbind(no_genre, main, interaction)

colnames(model_table) <- c("No Genre", "With Genre", "Interaction")

model_table |>
  round(3) |>
  knitr::kable() |>
  kableExtra::kable_styling(bootstrap_options = c("striped", "hover", "condensed"))
```

Table 1: Spotify. Comparison of Global Linear Models

	No Genre	With Genre	Interaction
r.squared	0.094	0.138	0.184
adj.r.squared	0.094	0.138	0.182
sigma	18.408	17.951	17.489
statistic	299.937	310.266	100.010
p.value	0.000	0.000	0.000
df	10.000	15.000	65.000
logLik	-125485.076	-124753.546	-123974.156
AIC	250994.153	249541.092	248082.311
BIC	251093.441	249681.750	248636.670
deviance	9813249.789	9329961.418	8841213.601
df.residual	28959.000	28954.000	28904.000
nobs	28970.000	28970.000	28970.000

1.4.2 Variable selection

Variable selection methods (`ols_step_backward`, `step`, and `drop1`), retained all predictors except speechiness, and the model was subsequently refitted.

```
lm_no_genre <- lm(popularity ~ . - genre, data = data)
lm_no_genre_no_speechiness <- lm(popularity ~ . - genre - speechiness, data = data)
```

```
model <- lm_no_genre_no_speechiness

# ols p-value method
model_ols <- {
  capture.output(tmp <- ols_step_backward_p(model))
  tmp
}

# AIC method
model_step     <- step(model, direction = "backward", trace = 0)

# drop method
model_drop <- {
  capture.output(tmp <- drop1(model, test = "F"))
  tmp
}


# Extract variables
attr(terms(model_ols$model), "term.labels")
```

```
[1] "duration_ms"      "danceability"     "energy"           "loudness"
[5] "acousticness"     "instrumentalness" "liveness"         "valence"
[9] "tempo"
```

```r
attr(terms(model_step), "term.labels")
```

```
[1] "duration_ms"      "danceability"     "energy"           "loudness"
[5] "acousticness"     "instrumentalness" "liveness"         "valence"
[9] "tempo"
```

```r
model_drop[!is.na(model_drop[, "Pr(>F)"]) &
           model_drop[, "Pr(>F)"] < 0.05, ]
```

	Df	Sum of Sq	RSS	AIC	F value	Pr(>F)
duration_ms	1	5853.683	9819437	168793.1	17.274287	0.0000324
danceability	1	199464.286	10013048	169358.8	588.621445	0.0000000
energy	1	100524.122	9914108	169071.1	296.647862	0.0000000
loudness	1	133021.750	9946605	169165.9	392.548743	0.0000000
acousticness	1	5770.813	9819354	168792.9	17.029738	0.0000369
instrumentalness	1	213296.224	10026880	169398.8	629.439657	0.0000000
liveness	1	35076.387	9848660	168879.2	103.510829	0.0000000
valence	1	21361.511	9834945	168838.8	63.038069	0.0000000
tempo	1	3127.167	9816711	168785.1	9.228306	0.0023851

1.4.3 Main effects

To ensure interpretability, analysis focuses on the no genre model, using numerical predictors only. The coefficient plot (Figure 1) shows that loudness and danceability have strong positive effects, whereas energy and instrumentalness have strong negative effects. Other variables have weaker negative effects.

```r
tidy(model, conf.int = TRUE) %>%
  filter(term != "(Intercept)") %>%
  filter(term != "genre") %>%
  ggplot(aes(x = estimate, y = reorder(term, estimate))) +
  geom_point(colour = "steelblue") +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.3) +
  geom_vline(xintercept = 0, linetype = "dashed", colour = "red") +
  theme_bw() +
  theme(axis.text.y = element_text(size = 7)) +
  labs(title = "Coefficient Plot (Main Effects Model)",
       x = "Estimate", y = NULL)
```

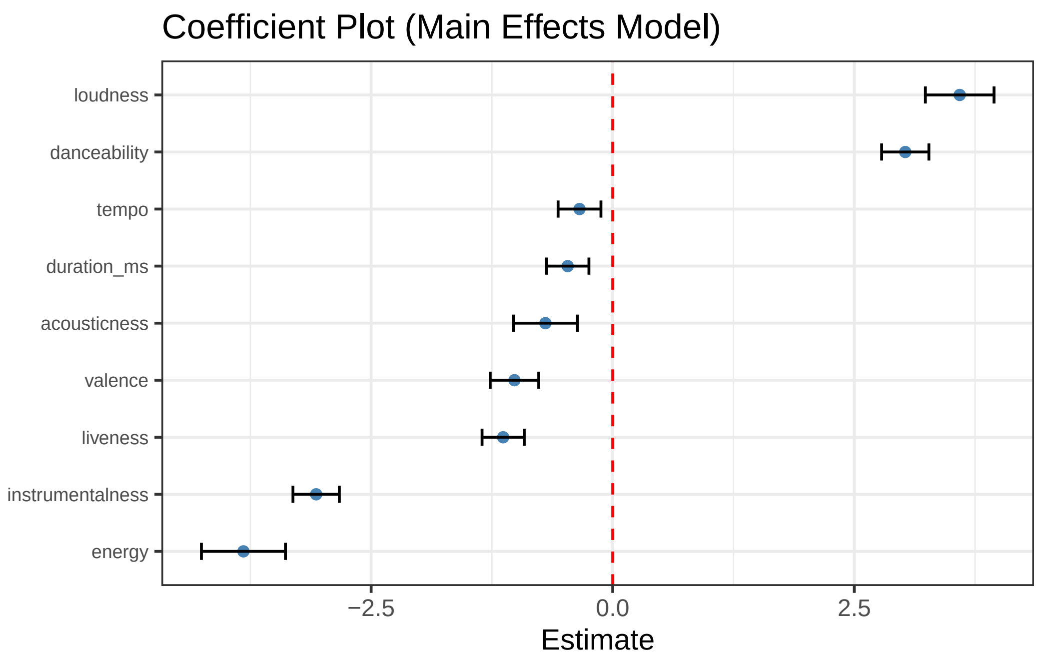

Figure 1: Spotify. Plot of the main effects coefficients

ANOVA results (Table 3, Figure 2) confirm this, with loudness and danceability having the largest F statistics, indicating the greatest explanatory power. Acousticness and tempo contribute minimally relative to other variables.

```
tidy(anova(model)) %>%
  filter(term != "Residuals") %>%
  ggplot(aes(x = statistic,
             y = reorder(term, statistic))) +
  geom_col(fill = "steelblue") +
  labs(title = "F-statistics from ANOVA table",
       x = "F statistic",
       y = NULL) +
  theme_bw()
```

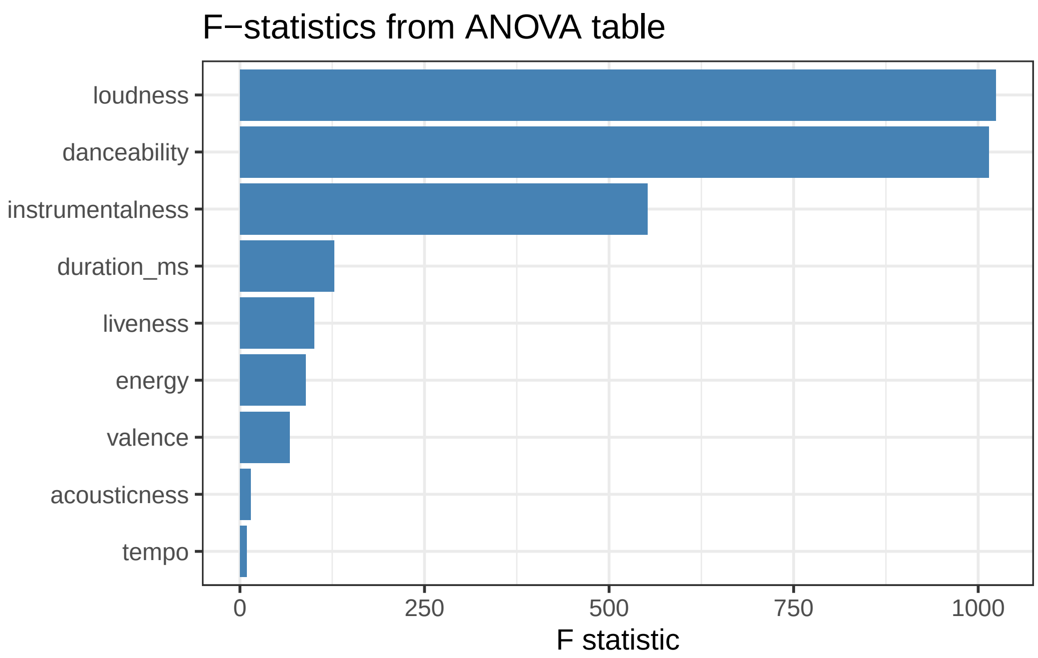

Figure 2: Spotify. Plot of F-statistics from ANOVA table

```
anova_model <- anova(model) |>
  knitr::kable(digits = 3)

anova_model
```

Table 3: Spotify. ANOVA table

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
duration_ms	1	43297.708	43297.708	127.772	0.000
danceability	1	343742.095	343742.095	1014.387	0.000
energy	1	30102.292	30102.292	88.832	0.000
loudness	1	346982.692	346982.692	1023.950	0.000
acousticness	1	5004.453	5004.453	14.768	0.000
instrumentalness	1	187026.426	187026.426	551.917	0.000
liveness	1	34047.486	34047.486	100.475	0.000
valence	1	22723.042	22723.042	67.056	0.000
tempo	1	3127.167	3127.167	9.228	0.002
Residuals	28960	9813583.546	338.867	NA	NA

1.4.4 Diagnostics

Diagnostics (Figure 3) indicate violations of normality and homoscedasticity. However, given the large sample size, these do not materially affect estimates, and the model is retained for interpretation rather than strict inference.

```r
par(mfrow = c(2, 2))
plot(model)
```

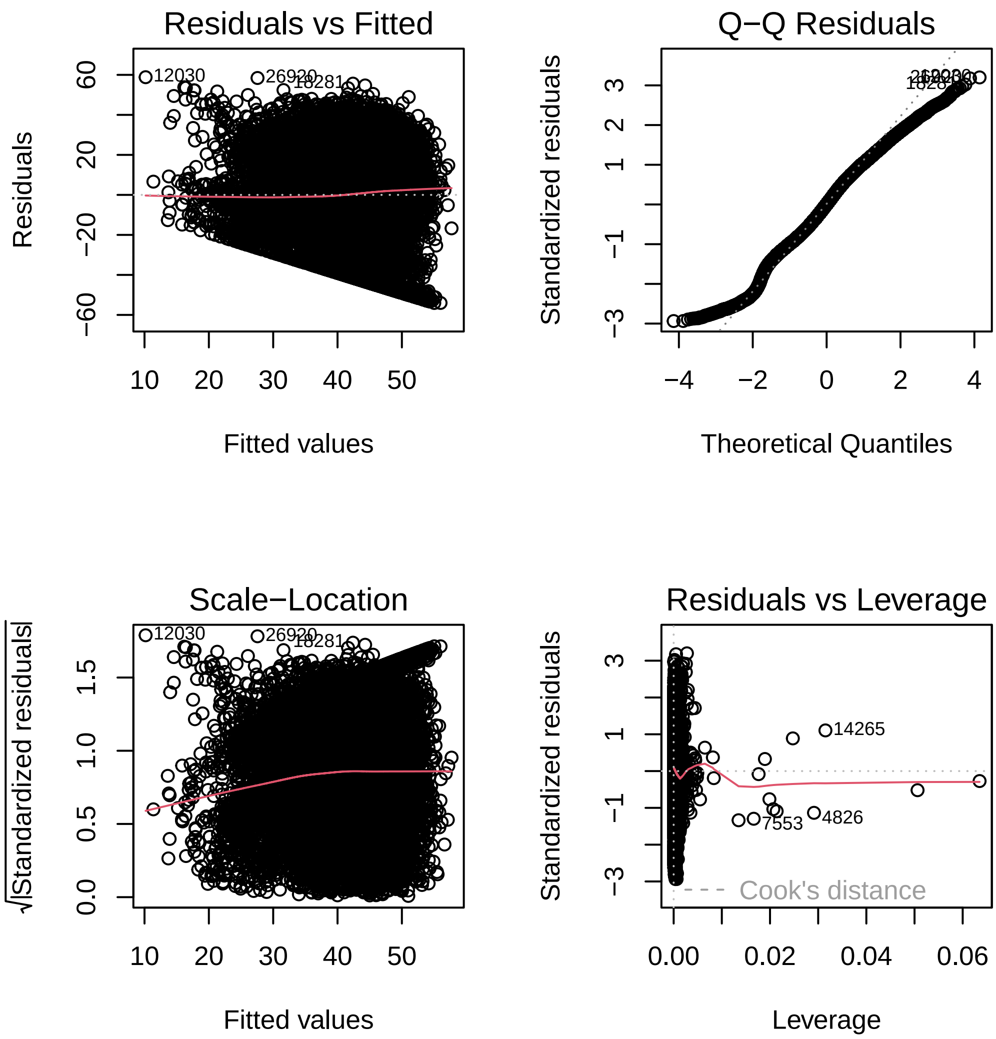

Figure 3: Spotify. Diagnostic plots

1.5 Part 2 - Genre-specific models

1.5.1 Model estimation

The data were split into genre-specific subsets, and seperate models fitted. Variable selection using `drop1`() was applied, and models were refitted using retained predictors.

Genre-specific models reveal both consistent and varying relationships (Figure 4). Danceability is positively associated with popularity across all genres (except not retained in HipHop), while liveness and valence (excl. Metal) were consistently negative. Loudness is positive in most genres but negative for Classical. These results align broadly with the no genre model (Figure 1).

In contrast, energy, instrumentalness, and duration vary in both magnitude and direction, indicating strong genre dependence and important heterogeneity.

```r
# genre sample size (n)
summary(spotify$genre)
```

```
 Classical Electronic     HipHop      Metal        Pop       Rock 
      1421       7508       1402       3927       7213       7499 
```

```r
# split into subsets
groups <- split(spotify, spotify$genre)

for (i in seq_along(groups)) {
  assign(paste0("genre_", i), groups[[i]])
}

# name subsets, drop genre
Classical  <- dplyr::select(groups[[1]], -genre)
Electronic <- dplyr::select(groups[[2]], -genre)
HipHop     <- dplyr::select(groups[[3]], -genre)
Metal      <- dplyr::select(groups[[4]], -genre)
Pop        <- dplyr::select(groups[[5]], -genre)
Rock       <- dplyr::select(groups[[6]], -genre)

# check subset dimensions
dim(Classical) # Classical (n=1421)
```

```
[1] 1421   11
```

```r
dim(Electronic) # Electronic (n=7508)
```

```
[1] 7508   11
```

```r
dim(HipHop) # HipHop (n=1402)
```

```
[1] 1402   11
```

```r
dim(Metal) # Metal (n=3927)
```

```
[1] 3927   11
```

```r
dim(Pop) # Pop (n=7213)
```

```
[1] 7213   11
```

```r
dim(Rock) # Rock (n=7499)
```

```
[1] 7499   11
```

```r
run_genre_model <- function(data) {

  # 1. Fit full model (no genre inside subset)
  model_full <- lm(popularity ~ ., data = data)

  # 2. Variable selection using drop1
  drop_tbl <- drop1(model_full, test = "F")

  # Convert to data frame for easier handling
  drop_df <- as.data.frame(drop_tbl)

  # Keep variables that are significant
  keep_vars <- rownames(drop_df)[
    !is.na(drop_df$`Pr(>F)`) & drop_df$`Pr(>F)` < 0.05
  ]

  # Remove "<none>" if present
  keep_vars <- setdiff(keep_vars, "<none>")

  # 3. Refit model with selected variables
  formula_new <- as.formula(
    paste("popularity ~", paste(keep_vars, collapse = " + "))
  )

  model_final <- lm(formula_new, data = data)

  # 4. Return everything useful
  list(
    full_model = model_full,
    selected_vars = keep_vars,
    final_model = model_final
  )
}
```

```r
res_classical  <- run_genre_model(Classical)
res_pop        <- run_genre_model(Pop)
res_rock       <- run_genre_model(Rock)
res_metal      <- run_genre_model(Metal)
res_hiphop     <- run_genre_model(HipHop)
res_electronic <- run_genre_model(Electronic)

# extract coefficients from each model
extract_coefs <- function(res, genre_name) {
  coefs <- summary(res$final_model)$coefficients
  df <- as.data.frame(coefs)
  df$Variable <- rownames(df)
  df$Genre <- genre_name
```

```r
  df
}

# combine
coef_all <- rbind(
  extract_coefs(res_classical, "Classical"),
  extract_coefs(res_pop, "Pop"),
  extract_coefs(res_rock, "Rock"),
  extract_coefs(res_metal, "Metal"),
  extract_coefs(res_hiphop, "HipHop"),
  extract_coefs(res_electronic, "Electronic")
)
```

```r
# plot
ggplot(coef_all[coef_all$Variable != "(Intercept)", ],
       aes(x = Genre, y = Estimate, fill = Genre)) +
  geom_col() +
  facet_wrap(~ Variable, scales = "free_y") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  theme_bw() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1)
  ) +
  labs(title = "Variation in Coefficients by Genre",
       x = NULL, y = "Estimate")
```

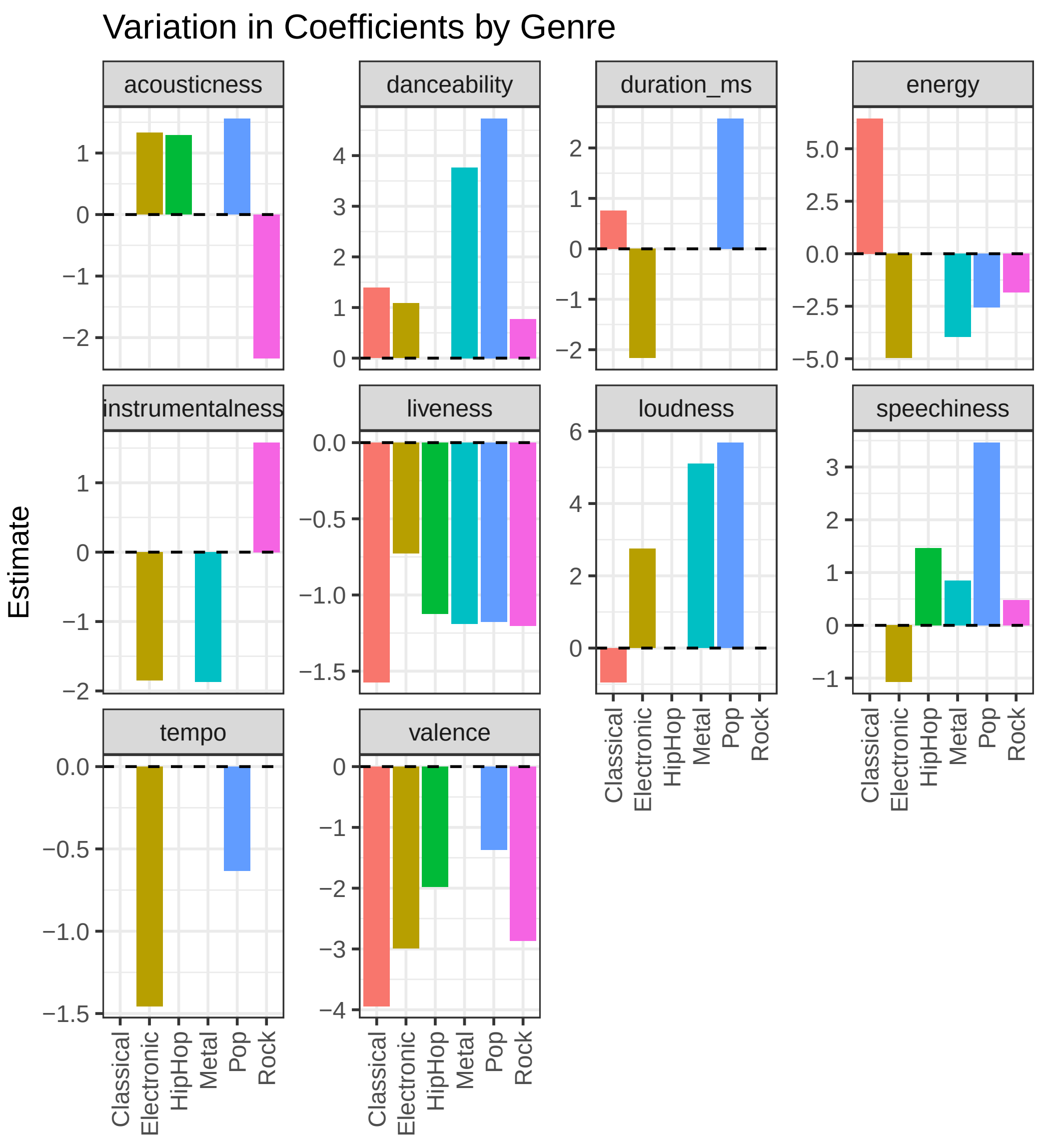

Figure 4: Spotify. Bar plots of coefficients by genre and audio variable

1.5.2 Cross validation

Cross-validation evaluates predictive performance (Table 4.), where lower values indicate better generalisation. Although genre-specific models have lower R^2, some achieve lower CV errors, suggesting improved prediction within genres. This highlights a trade-off between explanatory power and predictive accuracy.

```
compare_models_cv <- function(...) {
  models <- list(...)

  map_dfr(
    models,
    CV,
    .id = "model"
  ) |>
```

```r
  knitr::kable(digits = 3)
}
```

```r
compare_models_cv(
  No_Genre             = lm_no_genre,
  No_Genre_Speechiness = lm_no_genre_no_speechiness,
  With_Genre           = lm_main,
  Genre_Interaction    = lm_interaction,
  Pop                  = res_pop$final_model,
  Rock                 = res_rock$final_model,
  HipHop               = res_hiphop$final_model,
  Metal                = res_metal$final_model,
  Electronic           = res_electronic$final_model,
  Classical            = res_classical$final_model
)
```

Table 4: Spotify. CV summary table

model	CV	AIC	AICc	BIC	AdjR2
No_Genre	338.983	168780.854	168780.865	168880.142	0.094
No_Genre_Speechiness	338.970	168779.839	168779.848	168870.853	0.094
With_Genre	322.393	167327.793	167327.814	167468.452	0.138
Genre_Interaction	306.681	165869.012	165869.328	166423.371	0.182
Pop	322.907	41660.728	41660.765	41736.449	0.115
Rock	331.827	43532.405	43532.429	43594.707	0.044
HipHop	375.349	8312.993	8313.053	8344.467	0.025
Metal	223.664	21249.271	21249.308	21299.476	0.125
Electronic	315.678	43200.770	43200.811	43283.854	0.126
Classical	204.074	7559.464	7559.566	7601.537	0.131

1.6 Conclusions

1.6.1 Global model

Including genre improves model performance, with the adjusted R^2 accuracy increasing from approximately 0.09 to 0.18. Coefficient analysis shows that loudness and danceability have strong positive effects, whereas energy and instrumentalness have negative effects. ANOVA results confirm that loudness and danceability are the most important predictors. Despite this, overall explanatory power remains limited. Diagnostic checks indicate some assumption violations, but these are not critical given the large sample size.

1.6.2 Genre-specific models

Genre-specific models show both stable and varying effects. Danceability is consistently positive and liveness and valence consistently negative, while loudness positive in most genres indicating robust relationships. Other variables, including energy, instrumentalness, and duration, vary substantially, confirming that their effects depend on genre.

1.6.3 Limitations

The dataset appears artificially balanced across genres and does not include external factors such as artist popularity or marketing, limiting explanatory power. Some violations of model assumptions were observed, although these are mitigated by the large sample size. Additionally, grouping genres into broader categories may mask variation within subgenres. Tracks may also be associated with multiple genre labels in the original dataset; collapsing these to a single genre per track introduces a simplifying assumption that may not fully reflect the multi-genre nature of some songs.